if $\frac{d}{dx}(f(x)) < 0, \forall x \in (a,b)$ then $f(x)$ is strictly decreasing in the interval (a,b)

- A procedure for finding the absolute extrema of a continous function $f(x)$ on a closed interval $[a,b]$.

  Step 1 : Find the critical numbers of $f(x)$ in (a,b).

  Step 2 : Evaluate $f(x)$ at all critical numbers and at the endpoints a and b.

  Step 3 : The largest and the smallest of the values in Step 2 is the absolute maximum and absolute minimum of $f(x)$ respectively on the closed interval $[a,b]$.

- First Derivative Test

  Let $(c, f(c))$ be a critical point of function $f(x)$ that is continuous on an open interval I containing c. If $f(x)$ is differentiable on the interval, except possibly at c, then $f(c)$ can be classified as follows:(when moving across I from left to right)

  (i) If $f'(x)$ changes from negative to positive at c, then $f(x)$ has a local minimum $f(c)$.

  (ii) If $f'(x)$ changes from positive to negative at c, then $f(x)$ has a local maximum $f(c)$.

  (iii) If $f'(x)$ is positive on both sides of c, or negative on both sides of c then $f(x)$ has neither a local minimum nor a local minimum.

- Second Derivative Test

  Suppose that c is a critical point at which $f'(c) = 0$, that $f''(x)$ exists in a neighbourhood of c, and that $f'(c)$ exists. Then f has a relative maximum value at c if $f''(c) < 0$ and a relative minimum value at c if $f''(c) > 0$. If $f''(c) = 0$, the test is not informative.

ICT CORNER

https://ggbm.at/dy9kwgbt or Scan the QR Code

Open the Browser, type the URL Link given (or) Scan the QR code. GeoGebra work book named "**12th Standard Mathematics Vol-2**" will open. In the left side of work book there are chapters related to your text book. Click on the chapter named "**Applications of Differential Calculus**". You can see several work sheets related to the chapter. Go through all the work sheets.

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